Feng Tao 陶鳳, Ph.D.

Assistant Professor
College of Information Science and Technology
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EDUCATION

Ph.D. **Tsinghua University**, Department of Earth System Science, *with Honors*, Beijing, China, 2023
Joint Ph.D. at Max Planck Institute for Biogeochemistry, Jena, Germany (Sept 2021–Dec 2022)
Dissertation: *Mechanisms and Predictions of Global Soil Carbon Storage Based on Data Assimilation and Deep Learning*

B.Sc. **Sun Yat-sen University**, School of Environmental Science and Engineering, *with Honors*, Guangzhou, China, 2018
Exchange at The University of Hong Kong, Hong Kong SAR, China (Sept–Dec 2016)

ACADEMIC EXPERIENCE

2025– **The Pennsylvania State University**, University Park, PA, United States
Assistant Professor, College of Information Science and Technology
Institute of Energy and the Environment co-hire
Climate AI

2023–2025 **Cornell University**, Ithaca, NY, United States
Schmidt AI for Science Postdoctoral Fellow advised by Prof. Benjamin Z. Houlton, Department of Ecology & Evolutionary Biology
AI for scalable enhanced rock weathering to remove carbon dioxide

2018–2023 **Tsinghua University**, Beijing, China
Research Assistant of Profs. Yiqi Luo (at NAU/Cornell) & Xiaomeng Huang (at Tsinghua)
Soil carbon cycle study with data assimilation and deep learning

2021–2022 **Max Planck Institute for Biogeochemistry**, Jena, Germany
Joint Ph.D. student advised by Profs. Markus Reichstein & Marion Schrumpf
Global soil carbon sink using multi-source soil carbon ($^{12}C/^{14}C$) data

2019 **Food and Agriculture Organization of the United Nations**, Rome, Italy
Visiting Fellow of Global Soil Partnership (July–November)
Global soil organic carbon sequestration potential mapping

2017 **University of British Columbia**, Vancouver, Canada
Research Intern of Dr. Heather Trajano (June–September)
Microbial Fuel Cells (MFC)

RESEARCH INTERESTS

Soil organic carbon: formation, persistence, and responses to climate change

Enhanced rock weathering for scalable carbon dioxide removal

AI for mechanism discoveries in biogeochemistry (e.g., BINN: Biogeochemistry Informed Neural Network)

Process-guided and AI-assisted foundation model in biogeochemistry

GRANTS AND AWARDS

Grants

2023-2025 Schmidt AI for Science Postdoctoral Fellowship at Cornell University, on "Process-guided AI for scalable enhanced rock weathering toward global carbon neutrality", funded by Schmidt Sciences (USD180,000)

Awards

2025 Outstanding PhD Thesis in Beijing Region for the year of 2024

2023 27th “Xueshu Xinxiu” (Ten Outstanding Graduate Researchers) of Tsinghua University

2023 Honors PhD Graduate of Tsinghua University

2023 Outstanding PhD Thesis of Tsinghua University

2023 Best Student Paper Award of Sino-Ecologists Association Overseas (Sino-Eco) for paper *Microbial carbon use efficiency promotes global soil carbon storage* (Nature)

2021 Early Career ISMC (International Soil Modelling Consortium) Presentation Award

2018 Honors Bachelor Graduate of Sun Yat-sen University

2018 Outstanding Bachelor Thesis of Sun Yat-sen University

Scholarships

2020 Exchange Scholarship, funded by China Scholarship Council

2019 International Organization Internship, funded by China Scholarship Council

2017 Third Class Scholarship of Sun Yat-sen University

2017 Mitacs Globalink Research Internship Award, funded by Mitacs of Canada and China Scholarship Council

2017 Fung Scholarship (at HKU), funded by Victor and William Fung Foundation Limited

2016 First Class Scholarship of Sun Yat-sen University

2016 China National Scholarship, awarded by Ministry of Education of P. R. China

2015 First Class Scholarship of Sun Yat-sen University

2015 China National Scholarship, awarded by Ministry of Education of P. R. China

PUBLICATIONS

Google Scholar: https://scholar.google.com/citations?user=BlNzWiUAAAAJ

* Correspondence author

Work in Progress

Tao, F.* and B. Z. Houlton. A COUpled Soil Inorganic-organic carbon model for eNhanced wEathering (COUSINE) to manage carbon dioxide removal. ***under review***

Xu, H., J. Fan, **F. Tao***, C. Gomes, and Y. Luo. A Biogeochemistry-Informed Neural Network (BINN) for Mechanism Discoveries in Soil Carbon Cycle. ***under review***. Preprint available: https://doi.org/10.48550/arXiv.2502.00672

Tao, F.*, B. Z. Houlton, X. Huang, and Y. Luo. Deep soil carbon sequestration limited by low plant input. ***to be submitted***

Tao, F., H. Xu, Y. Luo, B. Z. Houlton, S. Shekhar, C. Gomes, J. Lehmann, Y. Sun, D. Woolf, D. Mulla, K. Pautian, V. Kumar, Z. Jin, S. Ogle, S. Pallickara, and S. Pallickara, Artificial intelligence to empower climate-smart agriculture and forestry for climate change solutions. ***to be submitted***

Selected Publications

2024 **Tao, F.***, B. Z. Houlton, Y. Huang, Y.-P. Wang, S. Manzoni, B. Ahrens, U. Mishra, L. Jiang, X. Huang, and Y. Luo. 2024. Convergence in simulating global soil organic carbon by structurally different models after data assimilation, ***Global Change Biology***, 30(5), e17297. https://doi.org/https://doi.org/10.1111/gcb.17297

2024 Huang, Y., X. Song, Y.-P. Wang, J. G. Canadell, Y. Luo, P. Ciais, A. Chen, S. Hong, Y. Wang, **F. Tao**, W. Li, Y. Xu, R. Mirzaeitalarposhti, H. Elbasiouny, I. Savin, D. Shchepashchenko, R. A. V. Rossel, D. S. Goll, J. Chang, B. Z. Houlton, H. Wu, F. Yang, X. Feng, Y. Chen, Y. Liu, S. Niu, and G.-L. Zhang. 2024. Size, distribution, and vulnerability of the global soil inorganic carbon. ***Science*** 384:233-239. https://doi.org/doi:10.1126/science.adi7918
Media Highlights: Phys.org, The Conversation, People's Daily

2023 **Tao, F.**, Y. Huang, B. A. Hungate, S. Manzoni, S. D. Frey, M. W. I. Schmidt, M. Reichstein, N. Carvalhais, P. Ciais, L. Jiang, J. Lehmann, Y.-P. Wang, B. Z. Houlton, B. Ahrens, U. Mishra, G. Hugelius, T. D. Hocking, X. Lu, Z. Shi, K. Viatkin, R. Vargas, Y. Yigini, C. Omuto, A. A. Malik, G. Peralta, R. Cuevas-Corona, L. E. Di Paolo, I. Luotto, C. Liao, Y.-S. Liang, V. S. Saynes, X. Huang, and Y. Luo. 2023. Microbial carbon use efficiency promotes global soil carbon storage. ***Nature*** 618, 981-985 (2023). https://doi.org/10.1038/s41586-023-06042-3
Media Highlights: Phys.org, TechnologyNetworks, Cornell Chronicle, VBiO (German), Forskning.se (Swedish), Science & Technology Daily (Chinese), Tsinghua News (Chinese)
Best Student Paper of Sino-Ecologists Association Overseas (Sino-Eco) 2023

2022 **Tao, F.**, and Y. Luo. PROcess-guided deep learning and DAta-driven modelling (PRODA). In Y. Luo & B. Smith (Eds.), *Land Carbon Cycle Modeling: Matrix Approach, Data Assimilation, and Ecological Forecasting*. Taylor and Francis.

Other Peer-reviewed Journal Articles

2025 Luo, Y., N. Wei, X. Lu, Y. Zhou, **F. Tao**, Q. Quan, C. Liao, L. Jiang, J. Xia, Y. Huang, S. Niu, X. Xu, Y. Sun, N. Zeng, C. Koven, L. Peng, S. Davis, P. Smith, F. You, Y. Jiang, L. Cheng and B. Houlton (2025). Large CO2 removal potential of woody debris preservation in managed forests. ***Nature Geoscience***. https://doi.org/10.1038/s41561-025-01731-2

2024 Nyaupane, K., U. Mishra, **F. Tao**, K. Yeo, W. J. Riley, F. M. Hoffman, and S. Gautam. 2024. Observational benchmarks inform representation of soil organic carbon dynamics in land surface models. ***Biogeosciences*** 21:5173-5183. https://doi.org/10.5194/bg-21-5173-2024

2024 He, X., E. Asb, S. Allison, **F. Tao**, Y. Huang, S. Manzoni, R. Abramoff, E. Bruni, S. Bowring, A. Chakrawal, P. Ciais, L. Elsgaard, P. Friedlingstein, K. Georgiou, G. Hugelius, L. B. Holm, W. Li, Y. Luo, G. Marmasse, N. Nunan, C. Qiu, S. Sitch, Y.-P. Wang, and D. Goll. 2024. Emerging multiscale insights on microbial carbon use efficiency in the land carbon cycle, ***Nature Communications***, 15(1), 8010. https://doi.org/10.1038/s41467-024-52160-5

2024 **Tao, F.**, J. Lehmann, Y.-P. Wang, L. Jiang, B. Ahrens, K. Viatkin, S. Manzoni, B. Z. Houlton, Y. Huang, X. Huang, and Y. Luo. 2024. Reply to "Beyond microbial carbon use efficiency", ***National Science Review***, Volume 11, Issue 4, April 2024, nwae058, https://doi.org/10.1093/nsr/nwae058

2024 **Tao, F.**, B. Z. Houlton, S. D. Frey, J. Lehmann, S. Manzoni, Y. Huang, L. Jiang, U. Mishra, B. A. Hungate, M. W. I. Schmidt, M. Reichstein, N. Carvalhais, P. Ciais, Y.-P. Wang, B. Ahrens, G. Hugelius, T. D. Hocking, X. Lu, Z. Shi, K. Viatkin, R. Vargas, Y. Yigini, C. Omuto, A. A. Malik, G. Peralta, R. Cuevas-Corona, L. E. Di Paolo, I. Luotto, C. Liao, Y.-S. Liang, V. S. Saynes, X. Huang, and Y. Luo. 2024 Reply to: Model uncertainty obscures major driver of soil carbon. ***Nature*** 627, E4-E6 (2024). https://doi.org/10.1038/s41586-023-07000-9

2024 **Tao, F.**, and B. Z. Houlton. 2024 Inorganic and organic synergies in enhanced weathering to promote carbon dioxide removal. ***Global change biology*** 30:e17132. https://doi.org/10.1111/gcb.17132

2023 Liao, C., X. Lu, Y. Huang, **F. Tao**, D. M. Lawrence, C. D. Koven, K. W. Oleson, W. R. Wieder, E. Kluzek, X. Huang, and Y. Luo. 2023. Matrix Approach to Accelerate Spin-Up of CLM5. ***Journal of Advances in Modeling Earth Systems***, 15(8), e2023MS003625. https://doi.org/https://doi.org/10.1029/2023MS003625

2023 Ma, S., L. Jiang, R. M. Wilson, J. Chanton, S. Niu, C. M. Iversen, A. Malhotra, J. Jiang, Y. Huang, X. Lu, Z. Shi, **F. Tao**, J. Liang, D. Ricciuto, P. J. Hanson, and Y. Luo. 2023. Thermal acclimation of plant photosynthesis and autotrophic respiration in a northern peatland. ***Environmental Research: Climate***, 2(2), 025003. https://doi.org/10.1088/2752-5295/acc67e

2023 Hou, E., S. Ma, Y. Huang, Y. Zhou, H.-S. Kim, E. López-Blanco, L. Jiang, J. Xia, **F. Tao**, C. Williams, M. Williams, D. Ricciuto, P. J. Hanson, and Y. Luo. 2023. Across-model spread and shrinking in predicting peatland carbon dynamics under global change. ***Global change biology*** 29:2759-2775.

2022 Liao, C., W. Huang, J. Wells, R. Zhao, K. Allen, E. Hou, X. Huang, H. Qiu, **F. Tao**, L. Jiang, M. Aguilos, L. Lin, X. Huang, and Y. Luo. 2022. Microbe-iron interactions control lignin decomposition in soil. ***Soil Biology and Biochemistry*** 173:108803.

2022 Luo, Y., Y. Huang, C. A. Sierra, J. Xia, A. Ahlström, Y. Chen, O. Hararuk, E. Hou, L. Jiang, C. Liao, X. Lu, Z. Shi, B. Smith, **F. Tao**, and Y.-P. Wang. 2022. Matrix approach to land carbon cycle modeling. ***Journal of Advances in Modeling Earth Systems***:e2022MS003008.

2022 Liao, C., W. Huang, J. Wells, R. Zhao, K. Allen, E. Hou, X. Huang, H. Qiu, **F. Tao**, L. Jiang, M. Aguilos, L. Lin, X. Huang, and Y. Luo. 2022. Microbe-iron interactions control lignin decomposition in soil. ***Soil Biology and Biochemistry*** 173:108803.

2021 Zhang, Z., H. Zhang, Z. Cui, **F. Tao**, Z. Chen, Y. Chang, V. Magliulo, G. Wohlfahrt, and D. Zhao. 2021. Global consistency in response of terrestrial ecosystem respiration to temperature. ***Agricultural and forest meteorology*** 308-309:108576.

2020 **Tao, F.**, Z. Zhou, Y. Huang, Q. Li, X. Lu, S. Ma, X. Huang, Y. Liang, G. Hugelius, L. Jiang, R. Doughty, Z. Ren, and Y. Luo. 2020. Deep Learning Optimizes Data-Driven Representation of Soil Organic Carbon in Earth System Model Over the Conterminous United States. ***Frontiers in Big Data***, 3(17). https://doi.org/10.3389/fdata.2020.00017

2018 Zhang, Z., W. Wang, J. Qi, H. Zhang, **F. Tao**, and R. Zhang. 2019. Priming effects of soil organic matter decomposition with addition of different carbon substrates. ***Journal of Soils and Sediments*** 19:1171-1178.

2018 Zhang, Z., R. Zhang, Y. Zhou, A. Cescatti, G. Wohlfahrt, M. Sun, H. Zhang, J. Qi, J. Zhu, V. Magliulo, **F. Tao**, and G. Chen. 2018. A temperature threshold to identify the driving climate forces of the respiratory process in terrestrial ecosystems. ***European Journal of Soil Biology*** 89:1-8.

Book Chapters

2022 **Tao, F.**. Bayesian statistics and Markov chain Monte Carlo method in data assimilation. In Y. Luo & B. Smith (Eds.), *Land Carbon Cycle Modeling: Matrix Approach, Data Assimilation, and Ecological*

Forecasting. Taylor and Francis.

2022 **Tao, F.**. Practice 10, Deep learning to optimize parametrization of CLM5. In Y. Luo & B. Smith (Eds.), *Land Carbon Cycle Modeling: Matrix Approach, Data Assimilation, and Ecological Forecasting*. Taylor and Francis.

PRESENTATIONS

Invited Talks/Seminars

2025 Woodwell Climate Research Center (invited by Dr Elchin Jafarov)

Yale Center for Natural Carbon Capture (YCNCC) Spring 2025 Symposium

Climate Change AI (CCAI) Discussion Seminar Series #6 (invited by Ivan Poon)
Recorded video available

2024 University of Zurich (invited by Prof. Michael W. I. Schmidt)

Institute of Earth Environment, Chinese Academy of Sciences (invited by Dr Ji Chen)

Sun Yat-sen University (invited by Dr Haicheng Zhang)

South China Botanical Garden, Chinese Academy of Sciences (invited by Dr Donghai Wu)

Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences (invited by Dr Yuanyuan Huang)

RUBISCO SOC WG Monthly Webinar (invited by Dr Umakant Mishra)

2023 Tennessee State University CCBB Journal Club (invited by Prof Jianwei Li)

Cornell University (invited by Prof. Fengqi You)

South China Botanical Garden, Chinese Academy of Sciences (invited by Dr Donghai Wu)

Sino-Eco

2022 RUBISCO SOC WG Monthly Webinar (invited by Dr Umakant Mishra)

Conference Presentations

2024 **Tao, F.** and B. Z. Houlton. Key limitations in enhanced rock weathering to promote carbon dioxide removal. 2024 AGU Fall Meeting, 9-13 December 2024, Washington DC, Poster.

2024 **Tao, F.** Y. Luo, X.Huang, and B. Z. Houlton. Deep soil carbon sequestration limited by low plant input. 2024 AGU Fall Meeting, 9-13 December 2024, Washington DC, Talk.

2024 **Tao, F.** and B. Z. Houlton. Coupling soil inorganic-organic carbon dynamics for verifiable carbon dioxide removal by enhanced weathering. 2024 Goldschmidt Conference, 18-23 August 2024, Chicago, IL, Talk.

2024 **Tao, F.** and B. Z. Houlton. A modeling approach to explore key mechanisms in enhanced weathering to promote carbon dioxide removal. 2024 ESA Annual Meeting, 4-9 August 2024, Long Beach, CA, Talk.

2024 **Tao, F.** and B. Z. Houlton. A Modeling Approach to Explore Inorganic and Organic Synergies in Enhanced Weathering to Promote Carbon Dioxide Removal. AOGS2024 21st Annual Meeting, 23-28 June 2024, Pyeongchang, Gangwon-do, South Korea, Talk.

2024 **Tao, F.** Convergence in simulating global soil organic carbon by structurally different models after data assimilation, 4th Annual Land Data Assimilation Community Virtual Workshop, 24-25 June, 2024, Virtual, Talk.

2023	**Tao, F.**, Y. Huang, B. A. Hungate, S. Manzoni, S. D. Frey, M. W. I. Schmidt, M. Reichstein, N. Carvalhais, P. Ciais, L. Jiang, J. Lehmann, Y.-P. Wang, B. Z. Houlton, B. Ahrens, U. Mishra, G. Hugelius, T. D. Hocking, X. Lu, Z. Shi, K. Viatkin, R. Vargas, Y. Yigini, C. Omuto, A. A. Malik, G. Peralta, R. Cuevas-Corona, L. E. Di Paolo, I. Luotto, C. Liao, Y.-S. Liang, V. S. Saynes, X. Huang, and Y. Luo. Microbial carbon use efficiency promotes global soil carbon storage. 2023 AGU Fall Meeting, 11-15 December 2023, San Francisco, CA, Poster.
2023	**Tao, F.**, B. Z. Houlton, Y. Huang, Y.-P. Wang, S. Manzoni, B. Ahrens, U. Mishra, L. Jiang, X. Huang, and Y. Luo. 2024. Convergence in simulating global soil organic carbon by structurally different models after data assimilation, 2023 AGU Fall Meeting, 11-15 December 2023, San Francisco, CA, Poster.
2022	**Tao, F.**, Ahrens, B., Yang, H., Schrumpf, M., Carvalhais, N., Reichstein, M., Huang, X. and Luo, Y., 2022. Historical fate of global soil organic carbon in the past century. 2022 AGU Fall Meeting. 12 - 16 December 2022, Chicago, USA, Talk.
2022	**Tao, F.** and Luo, Y., Quantifying soil carbon sequestration by multi-source constraints. 2022 EGU Meeting, 23 - 27 May 2022, Vienna, Austria, Talk (highlighted)
2021	**Tao, F.** and Luo, Y., Using Multi-source Constraints to Quantify Soil Carbon Sequestration. 2021 AGU Fall Meeting, 13 - 17 December 2021, Virtual, Poster
2021	**Tao, F.**, Huang, Y., Hungate, B., Lu, X., Hocking, T. D., Mishra, U., Hugelius, G., Huang, X., Luo, Y., PROcess-guided deep learning and DAta-driven modelling (PRODA) to uncover key patterns and mechanisms in global soil carbon cycle at OOS 25: Using Machine Learning to Quantify and Improve Earth System Predictions, 2021 ESA Annual Meeting, 2 - 6 August 2021, Virtual, Talk
2021	**Tao, F.** and Luo, Y., PROcess-guided deep learning and DAta-driven modelling (PRODA) uncovers key mechanisms underlying global soil carbon storage, 2021 3rd ISMC Conference, 18 - 22 May 2021, Virtual, Invited Talk, Highlights Talk
2021	**Tao, F.** and Luo, Y., Big data-driven modelling in CLM5 reveals microbial carbon use efficiency as the key mechanism underlying global soil organic carbon storage, 2021 NCAR CESM Land Model and Biogeochemistry Working Group Meeting, 23 - 25 February 2021, Virtual, Talk
2020	**Tao, F.**, Huang, X., Mishra, U., Hugelius, G. and Luo, Y., Big data-driven modelling reveals key mechanisms underlying soil organic carbon stabilization. 2020 AGU Fall Meeting, 1 - 17 December 2020, Virtual, Talk
2020	**Tao, F.**, Huang and Luo, Y., Deep learning and constrained modelling from big data jointly reveal key mechanisms in soil organic carbon stabilization at Symposium SYMP7: Combining Deep Learning and Process-Based Modeling to Advance Ecological Forecasting. 2020 ESA Annual Meeting, 3 - 6 August 2020, Virtual, Invited Talk
2020	**Tao, F.**. Improving representation of soil organic carbon: process-guided data driven deep learning modeling. 3rd Training Course on New Advances in Land Carbon Cycle Modeling, 21 - 30 July 2020, Virtual meeting hosted by Northern Arizona University, Flagstaff, Arizona, USA. Talk.
2019	**Tao, F.**., Zhou, Z., Huang, Y., Li, Q., Lu, X., Ma, S., Huang, X., Liang, Y., Hugelius, G., Jiang, L., Doughty, R., Ren, Z. and Luo, Y., Deep learning optimizes data-driven representation of soil organic carbon in Earth system model over the conterminous United States. 2019 AGU Fall Meeting, 9 - 13 December 2019, San Francisco, USA. Poster
2019	**Tao, F.**, Soil organic carbon sequestration potential: process-oriented model approach. 5th Working Session of the International Network of Soil Information Institutions (INSII), 21 - 23 October 2019, FAO Headquarters, Rome, Italy. Oral report on behalf of Global Soil Partnership (GSP) secretariat of FAO.

2018 **Tao, F.**, Luo, Y., Zhou, Z., Huang, Y., Big-data-big-model fusion to improve prediction of global soil carbon dynamics with Earth System Model. 2018 AGU Fall Meeting, 10 - 14 December 2018, Washington, D.C., USA. Poster

SERVICES

Guest Editor

2025 *Biogeochemistry*, Topical Collection: "Advances in land carbon cycle modeling". Leading Guest Editor with Drs Lifen Jiang, and Bailey Murphy.

2025 *Agriculture, Ecosystems & Environment*, Special Issue: "How agriculture managements influence biogeochemical cycles of soil carbon: processes, mechanisms and ecological consequences". Co-Guest Editor with Drs Xiaodong Song, Wenxiu Zou, and Jinsong Wang.

Academic Journal Peer Review

Review papers for *Nature Reviews Earth & Environment*, *Nature Communications*, *Global Change Biology* (GCB), *Soil Ecology Letters*, *GCB-Bioenergy*, *New Phytologist*, *Soil Biology & Biochemistry*, *Journal of Advances in Modeling Earth Systems* (JAMES), *Scientific Data*, *Communications Earth and Environment*, *Ecological Processes*, *Soil Science Society of America Journal*, *Catena*, *Soil Use and Management*

Other Review

Environment and Sustainability Honors Thesis Review, Cornell University (2024)

Conference Sessions

2024 **Tao, F.**, Y. Luo, Y. Sun, A. Bastos, and B. Murphy. Advances in land carbon cycle modeling, AGU Fall Meeting, 2024, Washington, D.C.

2024 **Tao, F.**, B. Z. Houlton, Y. Huang, Y. Luo, S. F. von Fromm. Soils in the Anthropocene, as a solution to climate change? Insights from empirical, modeling, and big data studies, AGU Fall Meeting, 2024, Washington, D.C.

2024 **Tao, F.**, B. Z. Houlton, and Y. Luo. Soil as a solution to climate change: insights from empirical, modeling, and big data studies, ESA Annual Meeting, 2024, Long Beach

Mentorship

2025– Kevin Toepelt, undergraduate student at Cornell, on transfer learning

2023– Haodi Xu, Ph.D. student at Cornell (primary supervisor: Yiqi Luo), on Biogeochemistry-Informed Neural Network (BINN)

2023– Joshua Fan, Ph.D. student at Cornell (primary supervisor: Carla P. Gomes), on Biogeochemistry-Informed Neural Network (BINN)

2024 Natalia Butler, Ph.D. student at Cornell, on ten-week rotation training on modeling enhanced rock weathering in Houlton Lab

TEACHING

Cornell University

2025 Graduate seminar (BioEE 7600-101, Spring, 1 credit): *AI in Ecology, Biodiversity, & the Environment*, as Co-Instructor with Drs. Felipe Pacheco, Imanol Miqueleiz, and Sebastian Heilpern

2024 Graduate seminar (BioEE 7600-102, Fall, 1 credit): *Global soil carbon cycle modeling and AI-aided new discoveries*, as Instructor

2022– Two-week Training Course: *New Advances in Land Carbon Cycle Modeling*, as Lecturer and Instructor

Northern Arizona University

2019–2021 Two-week Training Course: *New Advances in Land Carbon Cycle Modeling*, as Lecturer and Instructor

Tsinghua University

2021–2022 *Ecological Modelling*, as Guest Lecturer

ACADEMIC ADVISORS

Yiqi Luo (Ph.D., on Soil Carbon Science), Xiaomeng Huang (Ph.D., on Deep Learning), Benjamin Z. Houlton (Postdoc)

ADDITIONAL INFORMATION

Languages: Mandarin (Native), English (Proficient), German (Beginner)

Programming: Proficient at MATLAB, R and Python. Familiar with FORTRAN (MPI)